

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

25th June 2019

Subject SA4 – Pensions and Other Benefits

Time allowed: 3 Hours 15 Minutes (14.45 - 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which is meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the IRDA regulation. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the candidate should answer the question with reference to Indian environment.*
3. *Attempt all questions, beginning your answer to each question on a separate sheet.*
4. *Mark allocations are shown in brackets.*
5. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** An emerging country has introduced a voluntary defined contribution scheme for all citizens of the country. The key features of this plan are as follows:

Eligibility for entry	All citizens of the country aged 18 to 60
Enrolment	Voluntary
Contributions	Subscribers choose from the following contribution rates applicable for their whole membership: a) 250 per month b) 500 per month c) 1,000 per month d) 2,500 per month e) 5,000 per month
Exit	Only at retirement at age 65. No benefits are otherwise due to the subscriber before this date, except in the case of death where the total contributions paid are returned to a nominated beneficiary.
Benefit	At age 65, lump sum up to 40% of accumulated balance. Balance must be given to an insurance company to secure an annuity.
Investments	Each subscriber at the time of enrolment must choose from a lifecycle fund either: a) Aggressive where there is a higher equity portion (75% below age 35 grading down to 35% at age 55), rest is in Government bonds. b) Moderate where the equity portion is 50% below age 35 grading down to 20% at age 55, rest is in Government bonds. c) Conservative where the equity portion is 25% below age 35 grading down to 0% at age 55, rest is in Government bonds. Contributions are used to secure units in an equity pre-defined index tracking portfolio and units in a Government bond unit fund that tracks yields 10 year benchmark securities. Subscriber's balances are market related to the NAVs declared daily.
Fund managers	A subscriber must choose from one of the three empanelled fund managers. One can change fund manager twice throughout one's membership.
Charges	Subscribers pay 100 per annum for admin charges. Subscribers pay 5% of their first month contribution for opening expenses subject to a minimum of 100. NAV's are net of any investment management fees. The explicit fees are not disclosed separately.
Tax treatment	Contributions up to 12,000 per annum are tax permitted and can be used to reduce taxable income of the subscriber. All investment earnings are tax exempt. At retirement, lump sum is tax exempt to the extent of 20% of the accumulated balance. The balance given to the insurer for an annuity is not taxed but the annuity income is taxed as income in the hands of the subscriber.

Set out the points you would make in response to the following comments the Government has made to you:

- i) What are the risks of this scheme the subscribers should keep in mind? (3)

- ii)** What are the risks for the Regulator/Government? (2)
- iii) a)** Suggest areas where the design and features of the plan could be evolved to boost participation, in particular keeping a philosophy of flexibility and choice in mind. (5)
- b)** For each suggestion state key pros and cons for the subscriber and the Government. (10)

The Government wants to introduce a special scheme for Corporates to plug and play into the National Scheme on a voluntary basis.

- iv) a)** State at least 8 key areas the Government will need to consider when thinking about introducing this scheme. (6)
- b)** For each area, suggest two specific options that might be implemented. (4)

Besides above voluntary scheme, the country already has a mandatory defined benefit leaving service lump sum plan that employers have to provide for their employees. Currently, it is not pre funded by the employers and is operated on pay as you go basis. The Government is now proposing mandatory pre-funding by employers. The existing scheme is as follows:

- Mandatory for all employers to provide a leaving service benefit of one month salary for each year of service where an employer employs more than 10 employees.
- Payable after serving at least for 5 years with an employer.
- Payable on leaving service, voluntarily / death / retirement.

The proposal is that each employer must pre-fund the benefit by paying 4% of monthly salary to a Centralised Government Administrator. Employers will still be ultimately liable for the benefit obligation / payment and would receive reimbursement from their account held with the central administrator. Employers may apply for an exemption and run their own separate Trust.

- v)** State the key potential implications, considerations and questions for employers for the above proposal. (10)

Under the reporting rules for all companies, the Government has stated that an actuarial valuation at least every three years must be submitted to the Centralised Regulator to demonstrate that the Trust is funded sufficiently for the benefits stipulated.

- vi)** Comment on the considerations and options the regulator will have in setting the regulations for such a valuation. (10)
- [50]**

Q.2) “X” is a group of companies operating in India. It provides, as a part of employee benefit, defined pension benefit equivalent to 25% of final pay for those employees exiting after completing 25 years of service. In case an employee leaves before normal retirement age, the pension is provided from date of normal retirement or earlier death. The company is providing these benefits through an Independently Managed Trust and the funding is done by the Company based on recommendations of the Trustees. Over the period the funding status of the scheme shows increasing deficit and due to this, the contribution requirements to the Scheme as demanded by the Trustees are increasing. Concerned over the increasing pension cost, the Company has decided to take an independent review of the Scheme and you have been appointed as the Consultant Actuary to the Company X.

You have been given the following information about the Scheme during the initial meeting:

Year	2015-2016	2016-2017	2017-2018	2018-2019
Number of members (including pensioners)	10500	11300	11800	11900
Average age	42	42.6	43	44
Minimum pension (per month)	2000	2000	2500	3000
Normal retirement age (depending upon cadre)	60-62	60-62	59-61	59-61
PV of Benefit obligations (in million)	990	1050	1300	1375
Value of assets (in million)	900	920	930	950
Deficit (in million)	90	130	370	425

- i) List the possible sources of deficit for this Pension Scheme. (5)

During the initial meeting with the CFO of Company X, it has been decided to have further investigation into the finances of the scheme. You are preparing yourself for a meeting with the Trustees of the Scheme to be held shortly.

- ii) Outline the issues you propose to cover during the discussion with the Trustees and list the documents to be called from the Trustees. (14)

During the course of the investigation, you have noticed that there is a gradual strengthening of the basis used by Scheme Actuary especially the valuation rate of interest as under:

Year	2015-2016	2016-2017	2017-2018	2018-2019
Investment return earned on the assets	9.00%	8.75%	8.25%	8.00%
10 Year Government Bond Yield	8.25%	7.90%	7.80%	7.85%
Range of Discount rate used in Valuation	8.80% to 8.30%	8.50% to 7.90%	8.10% to 7.70%	7.85% to 7.70%

- iii) State the Professional Standards to be followed by you, before taking the assignment and while commenting on the work of the Scheme Actuary. (3)

- iv) Discuss the two approaches a) Asset Based approach, and b) Market consistent Approach in choosing the discount rates and discuss the reasons for the discount rates being different from the investment return and the 10 year Government bond Indices Yield. (8)

Your report to the Company has suggested several options available to the Company (& Trust) to manage the cost relating to the Scheme. One of the suggestion you have recommended is to use “insurance /pension products”.

- v) List the main 3 insurance/Pension group products available in Indian life Insurance market and state, for each such product, primary features, merits and demerits of using the product in managing the fund. (9)

In Indian market several insurers have started offering deferred annuity plans that provide guaranteed annuity pay-out from the vesting age. The Trustees of the Scheme are considering proposals of some of the insurers for purchasing the product to manage the pension liabilities.

Specifically they are considering the following proposals:

- a) Purchase of deferred annuities to meet the liabilities relating to deferred pensioners
 - b) Purchase of deferred annuities to meet the liabilities of the active members who are closer to retirement (i.e. relating to members aged 55 and above)
- vi)** Discuss the accounting treatment of each of the proposed purchases for the preparation of financials as per Ind AS19 and examine the impact of the proposals on Funding Status and pension cost.

(You may assume that such annuity purchases meet the definition of “Qualifying Insurance Policy” as given in the Accounting Standards and the pension Benefits are consistent with the annuity features).

(11)
[50]
